Estrogens and Progestins

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Block: Endocrine/Urology/Reproduction

Thread: Pharmacology

Conflict of Interest: Dr. Beck has no conflicts of interest to disclose related to this topic.

Textbook

Chrousos GP. Chapter 40 The Gonadal Hormones & Inhibitors. In: Vanderah TW. ed.
Katzung's Basic & Clinical Pharmacology, 16th Edition. McGraw-Hill; 2024. LINK

Levin ER, Vitek WS, Hammes SR. Estrogens, Progestins, and the Female Reproductive Tract. In:
Brunton LL, Knollmann BC. eds. *Goodman & Gilman's: The Pharmacological Basis of Therapeutics, 14e*. McGraw Hill; 2023.
LINK

Learning Objectives

By exam time, you should be able to:

1. Explain how the structure and function of the estrogen and progesterone receptors affect drug actions and tissue specificity.
2. Describe the mechanism of action, signature adverse effects, and contraindications of: fulvestrant, mifepristone; raloxifene, tamoxifen, ulipristal.

Key Words and Drugs

• Estrogen	• Tamoxifen	• Norethindrone
• Estrogen receptor	• Raloxifene	• Medroxyprogesterone
• Ethinyl estradiol	• SERD	• SPRM
• Mestranol	• Fulvestrant	• Ulipristal
• Anti-estrogens	• Progesterone	• Mifepristone
• SERM	• Progesterone receptor	

Study Questions

1. Discuss contraindications for postmenopausal HT (Hormone Therapy).
2. Compare and contrast the differences between tamoxifen and raloxifene in the treatment of hormone receptor positive breast cancer.

I. ESTROGEN AND PROGESTERONE – *hormones produced by the body*

A. Steroid structure

1. Steroids are lipids with a basic chemical structure of three 6-carbon rings and one 5-carbon rings.
2. Five classes of steroid hormones: estrogens, progestogens, androgens, mineralocorticoids, glucocorticoids

B. Metabolism/ Inactivation

1. Steroids are inactivated by conjugation to sulfate esters or glucuronic acids.
2. Metabolism: oxidation which occurs mainly, but not exclusively, in the liver.
3. Hormones are excreted in the urine.

C. Biosynthesis

1. Progesterone is the steroid hormone produced by the corpus luteum of the ovaries, the placenta, adrenal glands, and testis
2. Estrogens

a. produced primarily in the ovaries (both the follicle and corpus luteum) and the fetal-placenta unit.

b. Other sources:

Peripheral tissue (adipose tissue)- androstenedione is converted to estrone.

Liver –produces estradiol.

c. Estrogens are transported attached to proteins, primarily sex hormone binding globulin (SHBG) and

albumin. Only about 5% circulates unbound. SHBG has low capacity for binding but high affinity, albumin has high capacity but low affinity. Decreased SHBG leads to increased unbound circulating estrogens and androgens.

3. Neuroendocrine regulation of steroid production

a. Hypothalamic secretion of **GnRH** acts on pituitary to release gonadotropins

b. Anterior pituitary produces **FSH** (follicle stimulating hormone) and **LH** (luteinizing hormone) that act on the ovary and testis to stimulate steroid synthesis and gamete development.

D. Pharmacodynamics

1. Molecular mechanisms of steroid action

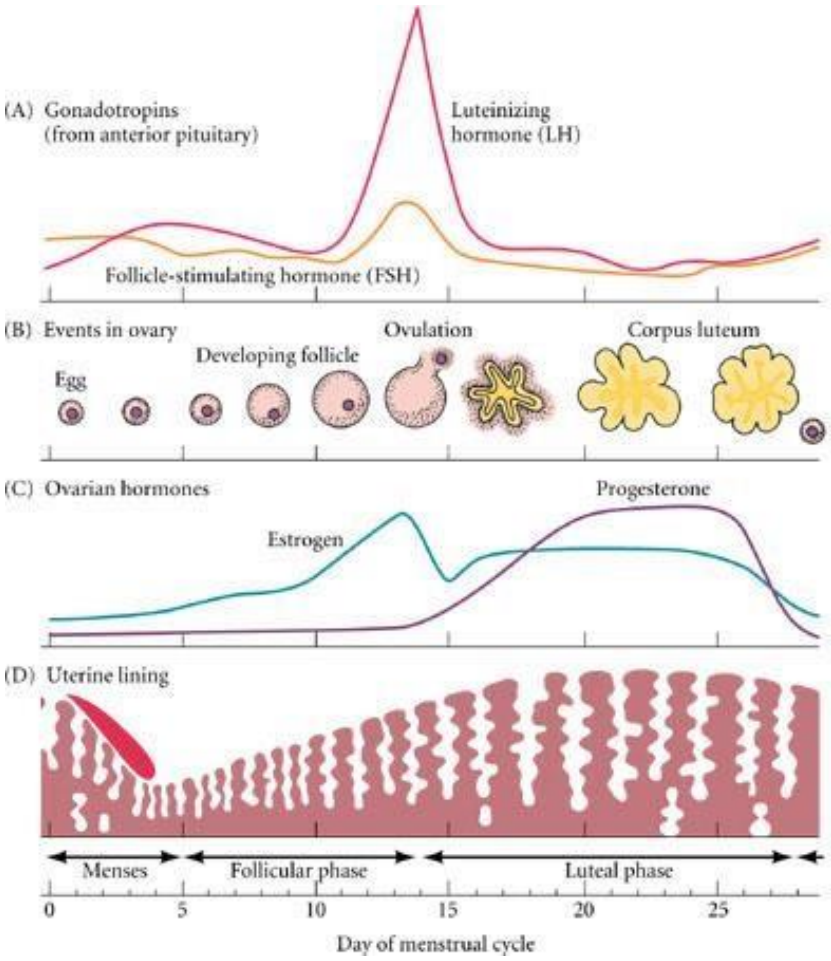
a. Steroid hormones enter most cells by diffusion

b. In target cells (i.e., cells sensitive to hormone), the steroid binds to receptors, which are located in the cytoplasm and/or the nucleus.

c. Steroid receptor is activated by hormone(ligand), dimerizes, binds DNA, induces transcription.

E. REVIEW:

Hypothalamic-pituitary-ovarian axis



F. REVIEW: Systemic estrogen effects (physiology)

Natal Female:

1. Anabolic effects

a. Promotes secondary sex characteristics at puberty, including pubertal growth spurt

2. Reproductive system

a. Regulation of menstrual cycle

b. Proliferation of uterine and vaginal epithelium

c. Stimulates production of cervical mucus, making it less viscous

Natal Male:

1. Stimulates growth of stromal cells in prostate
2. Essential for spermatogenesis

General:

2. Blood chemistry
a. Slight ↓ total cholesterol; slight ↑ TG
b. ↑ HDL and ↓ LDL
3. Suppresses cellular immune responses
4. Increases coagulation factors (increased coagulability) by:
   - ↓ antithrombin III
   - ↑ plasminogen, II, VII, IX, X
5. Bone: ↓ rate of bone resorption by ↓ osteoclast activity and number

G. REVIEW: Systemic progestogen/ progesterone effects (physiology)

1. Endocrine and reproductive system,
a. Regulation of the menstrual cycle
b. Modulates secretory differentiation and ↑ vascularity in uterus
c. Maintains pregnancy
d. Smooth muscle relaxant, represses uterine contractions

General:

1. Blood chemistry: ↓ HDL and ↑ LDL
2. Endocrine: effects on insulin levels and insulin response to glucose
3. Other:
   - body temperature

- breathing response to CO2
- Depressant/ hypnotic effect on brain

II. ESTROGEN and PROGESTERONE RECEPTORS

Both are nuclear receptors!

A. Estrogen receptors:

1. Human estrogen receptors alpha and beta isoforms have similar gene structure, but:

- different genes on different chromosomes
- different tissue/ cell distribution
- different affinity for ligands
- different genes activated

2. Receptor Activity

- ER in many body tissues
- ER-alpha and ER-beta levels depend on tissue
- ER-alpha more common in reproductive system
- ER-beta found in bone, brain, and urinary tract

3. Ligands for estrogen receptors

- Estrogens-estradiol
- Estradiol binds to ER-alpha and ER-beta
- Drugs designed to be selective for isoforms
  SERMS- Tamoxifen, Raloxifene, Toremifene
- Novel ER-alpha and ER-beta selective ligands

B. Progesterone Receptors

1. Progesterone Receptor has alpha and beta isoforms

- differential transcriptional activities
- cell type and promoter specific often PR beta > PR alpha
- PR alpha serves as repressor of PR beta and ER signaling

2. Ligands for Progesterone receptors

- Progesterones: progestins (example: norethindrone)

- Selective Progesterone Receptor Modulators aka SPRMS: ulipristal

C. ER/ PR Pharmacology/ Pharmacodynamics

There are two important interrelationships between estrogen and progestin receptor signaling pathways:

***in the uterus, PR opposes ER stimulation**

***in the breast, PR enhances ER stimulation**

- Estrogens and progestins act through ER and PR
- Ligands can be agonists, SERMs/SPRMs (mixed agonist antagonist), or antagonists

Selective Estrogen Receptor Modulator

Selective Progesterone Receptor Modulator

- Different ligands induce different receptor conformations that determine all further interactions

- ER and PR work with partner proteins that modulate their activity. Co-regulator levels vary in different tissues.
- Many interrelationships between ER and PR pathways
  a. progestins modulate estrogen action BUT also act independently
  b. progestin effects on estrogen actions vary in different tissues

III. Drugs acting on Estrogen and Progesterone Receptors

A. Selective ER modulators (SERMs) and Antagonists (SERDs)

Drugs that block ER activation and function are anti-estrogens.

What is the reason for the target site-specific effects of antiestrogens?

SERMs – Selective Estrogen Receptor Modulators:

Tamoxifen, Raloxifene **SERDs** – Selective Estrogen Receptor Down-Regulators: Fulvestrant

1. **Current hypothesis to explain the molecular mechanisms of antiestrogen action in different tissues:**
   a. PROTEIN CONFORMATION: antiestrogenic activity at the ER results in subtle changes in the whole protein confirmation that could be responsible d target sites.
   b. CO-ACTIVATORS FOR ER-

      Differences in the distribution of co-activators and co-repressors could contribute to tissue-specific differences in estrogenicity.
      1) activating function (AF-2) region in the ligand-binding domain is known to be repressed by tamoxifen and raloxifene, but the AF-1 domain is unaffected.
      2) Anti-ER complex may recruit completely new proteins at a specific target site to induce or to suppress gene transcription
   c. ALTERNATE RESPONSE ELEMENTS on DNA
      1) anti-ER complexes can bind to an estrogen response element but cannot recruit co-activators to initiate transcription of estrogen-responsive genes.

2. **Tamoxifen** (Nolvadex), a SERM-- widely used-- will discuss in detail
   a. Tamoxifen-is a non-steroidal triphenylethylene compound
   b. **MOA:** "Competitive partial agonist inhibitor of estradiol at the estrogen receptor"
   – Estrogen antagonist: effects centrally and on breast tissue
   – Estrogen agonist: effects on uterus and coagulation profile

   c. **Therapeutic Uses**

- effective in the treatment of metastatic breast cancer
- preventative of breast cancer in patients with high genetic risk
- adjuvant treatment of breast cancer
- ductal carcinoma *in situ*

Additional effects/benefits (but not primary reason to use):

- protects against bone loss
- decreases LDL cholesterol

d. **Adverse Effects**

- hot flashes
- vaginal bleeding
- endometrial hyperplasia/endometrial polyps

e. **Contraindications**

- patients who require warfarin- type anti-coagulation therapy
- history of deep vein thrombosis or pulmonary embolism

f. **Pharmacogenetics**

Polymorphisms in genes coding for tamoxifen metabolism may be predictive markers for response. (currently no pre-use testing required)

3. **Raloxifene** (Evista), a SERM

a. Exhibits tissue-selective estrogen antagonism/agonism.

b. MOA: "Partial estrogen agonist/ antagonist

Estrogen agonist activity in bone; estrogen antagonist activity in breast, uterine tissue

- Tamoxifen analogue (benzothiophene derivative). High binding affinity for ER
- Poor bioavailability due to rapid metabolism/ glucuronide conjugation; once daily dosing (despite t-1/2 of 32.5 hours)

c.Therapeutic uses:

- Postmenopausal osteoporosis prevention/ treatment: agonist activity in bone and on lipid metabolism (preserves bone mineral density, decreases LDL cholesterol and inhibits aortic cholesterol accumulation)
- antagonist activity on uterus, breast tissue (inhibits breast epithelium/uterine endometrium proliferation) May lower risk of breast cancer.
- No adverse effects on uterus.

d. **Contraindications**- history of deep vein thrombosis or pulmonary embolism

e. **Warnings:**
   - increased incidence of deep vein thrombosis and pulmonary embolism
   - no indication for pre-menopausal patients; appears to have opposite effect
   - may increase the incidence of hot flashes and is NOT effective in reducing hot flashes associated with estrogen deficiency.

4. Fulvestrant (Faslodex), an ER antagonist (also called a SERD)

a. **MOA:** estrogen receptor antagonist that binds to the estrogen receptor in a competitive manner with affinity comparable to that of estradiol

b. down-regulates the ER protein in human breast cancer cells.

c. **Indication:** treatment of hormone receptor positive metastatic breast cancer in postmenopausal patients with disease progression following antiestrogen therapy.

[NOTE: Clomiphene is a partial estrogen receptor agonist—it is a competitive inhibitor of endogenous estrogen (used to induce ovulation)—but not used in the treatment of breast cancer.]

B. Selective PR modulators (SPRMs) and Antagonists: partial agonist/ partial antagonist

1. **Ulipristal** (Ella), a SPRM

MOA: binds to the progesterone receptor; prevents ovulation.

Therapeutic uses: Emergency contraception

Effects depend on timing of the dose in the menstrual cycle.

Will NOT terminate an existing pregnancy.

2. **Mifepristone** (Mifeprex, RU-486), PR antagonist, cortisol receptor antagonist

MOA: competitive antagonist. Binds to progesterone receptors and inhibits activity of progesterone.

Therapeutic uses: Medical abortion; medical management of miscarriages; Cushing's syndrome

IV. Clinical Uses of Estrogens, Progestogens and their Antagonists

Summary of uses:

Estrogens as Therapeutic Agents

1. Therapeutically useful estrogens include: naturally derived and synthetic.

   a. Natural estrogens include estrone, estradiol, and conjugated equine estrogens.

   (Premarin = pregnant mare urine)

2. Estrogens are most often administered by mouth, but they can also be administered by multiple other routes.
3. Used for contraception, menstrual cycle control, ovulation induction, hormone therapy.

Progestogens as Therapeutic Agents

1. Prophylaxis against endometrial hyperplasia or carcinoma in estrogen treated women
2. In certain cases progestogen therapy is appropriate by itself:

a. induction of withdrawal bleeding in the diagnosis of amenorrhea

b. contraception

c. inhibition of gonadotropin secretion in girls with precocious puberty or in the treatment of endometriosis

A. Oral Contraception B. Other routes of administration for contraception C. Emergency Contraception D. Medical Abortion

This content (A-D) will be covered in the PHARM: Contraception & Abortion lecture and will not be covered in this lecture

E. Postmenopausal Hormone Therapy (HT)

Current practice: HT should not be used to improve cardiovascular health or treat osteoporosis, but an indication for vasomotor symptoms in a patient less than 10 years out from menopause is generally safe and can be used. Clinicians should discuss risks with patients at annual visit (risks vs. benefits/ quality of life issues)

The Women's Health Initiative (WHI) was a large randomized placebo-controlled trial evaluating HT in postmenopausal women. They found an increased risk of cardiovascular events with HT in the overall cohort, but secondary analyses showed that the risks were influenced by age and time since menopause. There were limitations to this study, but it had major impacts on utilization of HT and you will still see it cited and discussed.

Want to read a current review article summarizing the Issues?
Dr. Cullen recommends this article:
Levy B, Simon JA. A Contemporary View of Menopausal Hormone Therapy. *Obstet Gynecol.* 2024 Jul 1;144(1):12-23. PMID: 38484309.

1. **FDA–approved indications for HT**
    a. treatment of moderate to severe vasomotor symptoms
    b. treatment of moderate to severe atrophic vaginitis
    c. prevention of postmenopausal osteoporosis, NOT treatment
2. **Estrogens:** Various forms and administration routes used (vaginal creams, inserts/rings can limit systemic exposure)
3. **Rationale for progestin use**
    a. Primary menopause-related indication for progestin use is endometrial protection from unopposed estrogen therapy
    b. Endometrial cancer risk increases 4-20 times with unopposed estrogen (ET); adding progestogen (EPT) attenuates risk
    c. Continuous-combined EPT or continuous-sequential EPT regimens are appropriate
4. **Guiding Principles for ET/EPT therapy**
    a. Lowest effective dose
    b. Shortest duration required
    c. Type/route/regimen of E and P

d. Individualized risks and benefits
e. Patient satisfaction with therapy

5. **Contraindications to ET/EPT**
   a. Hormone-sensitive cancer
   b. Unexplained uterine bleeding
   c. Active liver disease (esp. with oral ET)
   d. History of deep venous thrombosis/ pulmonary embolism (DVT/PE)
   e. Confirmed cardiovascular disease (CVD)
6. **Adverse effects**
   a. Uterine bleeding
   b. Breast tenderness
   c. Abdominal bloating

Learn more: Practice Bulletin Number 141, January 2014 (reaffirmed 2018), Management of Menopausal Symptoms, American College of Obstetricians and Gynecologists.

F. Breast cancer therapeutics

You have had a CBL case of a patient with breast cancer.

G. Osteoporosis treatment and prophylaxis

You will have a lecture on osteoporosis pharmacology in Block MSK

Summary

- Review of estrogen and progesterone physiology in context of understanding drug action and adverse effects
- Discussion of estrogen and progesterone receptors and tissue specificity of isoforms
- Drugs acting on estrogen and progesterone receptors: SERMs, SERDs, SPRMs
- Clinical uses (discussion of HT)

Answers to Study Questions

These answers are not meant to be all encompassing; rather they are examples of the types of thought processes needed for application of the material.

1. Discuss contraindications for postmenopausal HT (Hormone Therapy).

   Answer: Contraindications to postmenopausal HT include

   i. *a history of breast cancer*
   ii. *coronary heart disease*
   iii. *a previous venous thromboembolic event or stroke*
   iv. *transient ischemic attacks*
   v. *active liver disease*
   vi. *unexplained vaginal bleeding*
   vii. *high-risk endometrial cancer*

2. Compare and contrast the differences between tamoxifen and raloxifene in the treatment of hormone receptor positive breast cancer.

   Answer

   Same:

   - *Both are selective estrogen receptor modulators.*
   - *Both are FDA-approved for treatment of hormone receptor positive breast cancer.*
   - *Both can cause hot flashes as adverse effects.*
   - *Both can improve osteoporosis because both act as ER agonists on bone (but only raloxifene has an indication for this)*
   - *Raloxifene is as effective as tamoxifen in reducing the risk of invasive breast cancer*
   - *Both tamoxifen and raloxifene can increase the risk of deep venous thrombosis.*

   Different:

   - *Tamoxifen acts as an ER antagonist on the breast but is an ER agonist on the endometrium/uterus.*
   - *Raloxifene is an ER agonist on both breast and endometrium/uterus ER.*
   - *Risk of endometrial cancer in raloxifene users is less than that of nonusers.*
   - *Tamoxifen users have a risk of developing endometrial cancer three times higher than that of raloxifene users (particularly if age over 50)*